🗂️ Project Task Plan: **Product Management Feature Development**

☑️ Step 1: Create “Add Product” Component in Admin Panel

- Admin dashboard me ek naya component banaya jayega jiska naam **AddProduct** hoga.
- Yeh component **admin-specific route** par accessible hoga.
- Component ka purpose hoga: naye products ko form ke zariye add karna.

☑️ Step 2: Build Product Form UI with Design

- AddProduct component ke andar ek **structured product form** banaya jayega.
- Form me following fields honge:
  - Product Title
  - Description
  - Category
  - Price
  - Discount (optional)
  - Stock
  - Product Image Upload
- Form ke saath proper **UI/UX design** implement kiya jayega using CSS/Frameworks (Tailwind).
- Input validations aur placeholders bhi form me rahenge for better UX.

☑️ Step 3: Create Product Model in Backend

- Backend me ek naya model banega: `Product.js`
- Model ke andar wo sabhi fields define honge jo form me diye gaye hain.
- Fields ka data type, required validation, aur default values backend level pe handle honge.

☑️ Step 4: Develop POST API to Add Products

- Controllers folder me ek nayi function banegi: `addProductController`
- Ek POST API route define hoga: `POST /api/products/add`
- Yeh API client ke form se aayi values ko `req.body` se read karegi.
- Backend validation bhi hoga (e.g., title required, price must be number).
- Product image agar file me upload ho rahi hai, to file handling logic bhi add hoga (Multer or Cloudinary etc.)
- Response me success ya error ka message return hoga.

☑ Step 5: Connect POST API with Frontend Form

- Jab POST API postman pe successfully test ho jaye, tab is API ko frontend form ke saath integrate kiya jayega.
- Form submit hone par `handleSubmit()` method call hoga jo API call karega using **Axios** or **Fetch**.
- Submit ke baad success message ya error message UI me show hoga.
- Optionally, form clear ho sakta hai after submission.

☑ Step 6: Create GET API to Fetch All Products

- Ek GET API banayi jayegi: `GET /api/products`
- Yeh API database se sabhi products ko fetch karegi.
- Filters (like category, price range) future enhancement ke liye optional rahenge.
- Response me ek array return hoga with all products.

☑ Step 7: Render Products in User Panel

- User panel (Home Page ya Product Page) me `useEffect` ya similar method se GET API call hogi.
- API se aaye data ko UI me map karke **Product Card Component** me show kiya jayega.
- Product card me title, image, price, category, aur optional fields show honge.
- Design responsive aur user-friendly banaya jayega.

☑ Final Goal:

- Admin new products add kar sake.
- Users ko sabhi products show ho jayein structured card format me.
- Proper backend validations aur API structure maintained ho.
- Full stack flow: **Frontend Form → Backend POST API → DB → GET API → Frontend Rendering**